SVKM's NMIMS · Mukesh Patel School of Technology Management & Engineering, Mumbai

Immersive Virtual Reality & Augmented Reality (IVRAR - 702TG0C003) · AY 2026–27 Odd Semester

GROUP 08 TECHNICAL DOSSIER

Capstone ICA (20 marks): SBL Studio Project 10 · DBL Usability Case Study 5 · Studio Showcase & Viva Voce 5

Group / Batch	Group 08 · Batch B2 (as allotted in the finalised PBL titles sheet)
Monorepo folder	https://github.com/sunny-nanade/NMIMS-MPSTME-IVRAR-2026/tree/main/Group_08_Varun_Arjun_Om
Submit (3 places)	(1) PDF on MS Teams, turned in by EVERY member · (2) this filled Word file AND its PDF committed to Group_08_Varun_Arjun_Om/docs/IVRAR_Group_08_Technical_Dossier.docx / .pdf · (3) LinkedIn video URL in Section 5. Due Monday, 26-Oct-2026, 23:59 IST. Studio Showcase, Expo & Viva Voce: Saturday, 31-Oct-2026.
How to fill this file	Use Microsoft Word (desktop recommended). Choose answers from the 'Select ▾' drop-downs and type into the 'type here' fields. Do not delete or retype them, and do not edit this file in Google Docs: the drop-downs are how your answers are read.

Student name	Roll No.	SAP ID	Programme	Batch	Declared studio role
VARUN IYER	F050	70372200014	B.Tech (Info Technology) Integrated	B2	Mobile AR Lead
ARJUN SALUNKE	F049	70372200015	B.Tech (Info Technology) Integrated	B2	XR Systems Architect
OM KADAM	F014	70372200021	B.Tech (Info Technology) Integrated	B2	Gamification & Telemetry Specialist

Batch column = official student list. Tell Dr. Nanade by 20-Oct if any name, ID or role is wrong.

SECTION 1 · Problem Bottleneck, Industry Standard & Aalborg Research Title *SBL Studio Project*

Authorised Aalborg title	To what extent does a gamified mobile AR checkpoint discovery system enhance campus orientation and navigational self-efficacy among incoming university students?
Original draft title	How can an AR/VR-based campus orientation system improve first-year student engagement with MPSTME academic and campus facilities within the first 2 weeks, where institutional observations indicate that many incoming students face difficulty navigating unfamiliar facilities and accessing key academic resources during their initial weeks?
Why the title changed	The submitted title ('MPSTME Treasure Hunt') sounded like an informal student fest activity. Lacked theoretical rigor and measurable cognitive outcomes.

1.1 Baseline friction & governing standard: claim verification table

Each statement below was carried over from your approved title sheet. Before you use it anywhere (dossier, video, paper), find the primary source yourself and record it, or correct or delete the claim. Standards: cite number, part and year. Statistics: cite the original study, not a blog or an AI answer.

Claim (as in title sheet)	Faculty pre-audit note	Primary source you verified (author, year, venue, DOI/URL)	Source type	Keep / Correct / Delete
Bandura's Perceived Self-Efficacy applied to spatial navigation.	*Self-Efficacy: Toward a Unifying Theory of Behavioral Change*	*Psychological Review, 1977, 84(2), 191–215, DOI 10.1037/0033-295X.84.2.191*	Official report / documentation	Keep
Keller's ARCS (Attention, Relevance, Confidence, Satisfaction) model.	*Development and Use of the ARCS Model of Instructional Design*	*Journal of Instructional Development, 1987, 10, 2–10, DOI 10.1007/BF02905780*	Official report / documentation	Keep
Mobile GPS/Image anchor accuracy in institutional environments.	*The desktop prototype uses keyboard movement, raycast-based interaction, and collider-triggered facility visits*	Own Code	No source found	Keep

1.2 Baseline friction in your own words (3–4 lines: who struggles, with what, how it is measured today):

C# scripts in your monorepo folder today	ARCheckpointDiscoveryManager.cs · NavigationalSelfEfficacyLogger.cs
Telemetry logger named in the title sheet	GamifiedDiscoveryLogger.cs
Core state machine	Locked checkpoint → Within range → Image anchor found → Badge unlocked → Next checkpoint
Fundamental equations	Completion Time = checkpoints found in time Self-efficacy change: ΔSE = SE_post − SE_pre
Toolchain (title sheet)	Unity 2022.3 LTS; AR Foundation (ARCore/ARKit); 3D unlockable badge assets; GPS location triggers; C# GamifiedDiscoveryLogger.cs.

2.1 Authorship of each script

Several scripts were provided as starter scaffolds. Be honest: mark what you wrote, what you changed, and what you used as given. Viva questions will target the code you claim.

Script	Written / Modified / Used as given	Main author (Roll)	What you changed or added (1 line)
ARCheckpointDiscoveryManager.cs	Modified by us	Shared	--
NavigationalSelfEfficacyLogger.cs	Modified by us	Shared	--
MissionSetup.cs	Written by us	Shared	--
CanteenOrderingTests.cs	Written by us	Shared	--

2.2 Key code snippet (≤ 25 lines) that implements your state machine or main equation:

```
public static void Run(){
 checks=0;var go=new GameObject("Mission test fixture"){hideFlags=HideFlags.HideAndDontSave};
 var facility=ScriptableObject.CreateInstance<FacilityData>();var q=ScriptableObject.CreateInstance<QuizQuestionData>();var
rules=ScriptableObject.CreateInstance<ScoringRules>();
 try{
 var score=go.AddComponent<ScoreManager>();score.rules=rules;
 Check(score.Award("facility:a",50)&&score.Total==50,"first facility reward");
 Check(!score.Award("facility:a",50)&&score.Total==50,"duplicate facility reward rejected");
 score.Penalty(-10);Check(score.Total==40,"wrong-floor penalty");score.Penalty(5);Check(score.Total==40,"positive penalty cannot reward");
 score.Award("invalid",-100);Check(score.Total==40,"negative award
clamped");score.ResetSession();Check(score.Total==0&&score.Award("facility:a",50),"restart clears awards");
 var cp=go.AddComponent<CheckpointManager>();Check(cp.Mark("stairs:G-1")&&cp.Has("stairs:G-1"),"stair checkpoint");Check(!cp.Mark("stairs:G-
1"),"duplicate checkpoint");Check(!cp.Has("lift:7"),"unvisited route not granted");cp.ResetSession();Check(!cp.Has("stairs:G-1"),"checkpoint reset");
 var timer=go.AddComponent<GameTimer>();timer.limitSeconds=5000;timer.Tick(1199);Check(timer.Remaining==1,"20-minute
cap");timer.Tick(3);Check(timer.Remaining==0,"timer expires");timer.ResetSession();Check(timer.Elapsed==0,"timer reset");
 facility.facilityID="test";facility.isVerified=true;facility.isPlaceholder=false;
 q.facility=facility;q.isVerified=true;q.answers=new[]{"A","B"};q.correctIndex=0;
 Check(q.Eligible,"verified question accepted");q.isPlaceholder=true;Check(!q.Eligible,"placeholder question excluded");q.isPlaceholder=false;
 facility.isPlaceholder=true;Check(!q.Eligible,"placeholder facility
excluded");facility.isPlaceholder=false;facility.isVerified=false;Check(!q.Eligible,"unverified facility excluded");facility.isVerified=true;
```

2.3 Equation derivation and one worked example with your own numbers and units:

- ***Mission completion*** *= completed missions ÷ 7 × 100%.*
- ***Score*** *= mission completion awards + facility awards + correct quiz awards + no-mistake route awards + penalties.*
- *The configured point values are +100 per completed mission, +50 per facility, +50 per correct quiz, +25 for a mission completed without mistakes, −10 for a wrong-floor event, and −5 for a wrong quiz answer.*
- ***Illustrative calculation only:*** *one completed mission (+100), one new facility (+50), one correct quiz answer (+50), and a mistake-free route (+25), with one wrong-floor penalty (−10), gives* ***215 points****. This is not a participant result.*

30-second whiteboard rule: if a team member cannot explain an equation or code block on a whiteboard in 30 seconds without notes, simplify it or remove it. It will be asked in the viva.

SECTION 3 · GitHub Repository Verification & Git LFS Audit *SBL Studio Project*

Repository (monorepo)	https://github.com/sunny-nanade/NMIMS-MPSTME-IVRAR-2026
Your group folder	Group_08_Varun_Arjun_Om/
Filled dossier location	Group_08_Varun_Arjun_Om/docs/IVRAR_Group_08_Technical_Dossier.docx and .pdf
Release tag	group-08-v1.0 · build: Android .apk (ARCore)

Final commit hash (full 40 characters)	type here
Release build link (.apk / .exe)	type here
Root .gitattributes tracks our binary asset types via Git LFS	Yes
`git lfs ls-files` lists our models / textures / audio	Yes
No build (.apk / .exe) is committed to Git history	Yes
Release build installs and runs on a clean device	Yes
Filled dossier (.docx + .pdf) committed to docs/	Yes

3.1 Repository telemetry for your folder (run from the repo root; type the numbers)

Metric	Command	Value
Commits touching your folder	git log --oneline -- Group_08_Varun_Arjun_Om/ | wc -l	14
Branches used by your group	git branch -r --list '*group-08*' | wc -l	2
Lines of C# code	find Group_08_Varun_Arjun_Om -name '*.cs' | xargs cat | wc -l	1918
Total folder size (MB)	du -sm Group_08_Varun_Arjun_Om	1971
LFS-tracked files in your folder	git lfs ls-files | grep Group_08_Varun_Arjun_Om | wc -l	--
Raw telemetry CSV files (real participants)	ls Group_08_Varun_Arjun_Om/telemetry/*.csv | wc -l	--

3.2 Individual contribution (from git history; this table is checked against GitHub)

Member	GitHub username	Commits	% effort (agreed)	Main contribution (files / features)
Varun Iyer (F050)	Varun-210806	4	33%	Meta XR Setup, Mobile AR Lead
Arjun Salunke (F049)	Arjunnn143.2	5	33%	Game Logic, Code debugging, Functionality Testing, etc
Om Kadam (F014)	Om-Kadam	5	33%	World Building, Scene Setup, Game setting, etc

SECTION 4 · Human-in-the-Loop Usability Evaluation & Empirical Data Pack *DBL Usability Case Study*

Primary task metric	Completion Time(Mins)
Participants	Minimum N ≥ 10 real people (your PBL dossier recommends N = 24). Voluntary, adult, written consent; anonymised IDs (P01, P02…).
Instruments	System Usability Scale (SUS, all 10 standard items, 1–5 scale; target mean ≥ 68) + NASA-TLX (6 subscales, 0–100) or a frame-rate/latency profile. Report Simulator Sickness (SSQ) if users wear a headset.
SUS scoring	Odd items: score − 1. Even items: 5 − score. Sum × 2.5 = SUS (0–100) per participant.

Real participants tested (N)	10
Study design	Within-subjects (everyone tries both)
Testing dates (from – to) and place	29-10-2026 to 30-10-2026
Raw CSV path in repo	type here

Measure	N	Mean	SD	Median	Condition / comparison
SUS score	NA	NA	NA	NA	NA
NASA-TLX (raw)	NA	NA	NA	NA	NA
Primary task metric	10	11.5 min	1 min	11 min	Time to Complete Missions

[SUS box-plot (per condition)]	*[Task-metric / latency plot]*

DATA PROVENANCE DECLARATION (mandatory). Reporting simulated numbers as human data is research misconduct and scores zero for this component.

Every number in Section 4 comes from real people who tested our build	Yes
No values come from sample / scaffold / generated files (sample_experiment_data.csv, *_benchmark.csv, generate_paper_figures.py)	No
Every participant signed a consent form and we hold the forms	No
Our plots re-generate from the raw CSV in the repo	No

SECTION 5 · LinkedIn Video Showcase, AI Verification Audit & Peer Review *Studio Showcase & Viva Voce*

LinkedIn post URL	https://www.linkedin.com/posts/arjun-salunke-438117323_nmims-mpstme-virtualreality-ugcPost-7522217237605138432-pgcH/?utm_source=social_share_send&utm_medium=ios_app&rcm=ACoAAFG1fqwBp7QfXv7BAK0LemtcHNsjLmwGrP0&utm_campaign=copy_link
Video length (mm:ss)	03:07
Every member appears and speaks in the video	Yes

5.1 AI Interaction Audit Log: three cases where AI was wrong, made something up, or was unverifiable

Examples from this course: a fake 'Verizon 2024 XR' report, a fictional 'NEXORA Corp', a correct paper given the wrong journal. Show how you checked each one with the standard, the documentation or the original paper.

#	AI tool	What we asked & what AI said	How we caught it	What was actually true (with source)	Error type	Verdict
AI-01	ChatGPT	Scoring System Setup – Gave unrealistic point schemes and overcomplicated the game	Ran / tested the code	Simpler math for scoring and simpler scoring and calculation	Wrong number or statistic	Corrected
AI-02	ChatGPT	Timing Constraints – Time Constraints were too difficult for first time players	Ran / tested the code	Made it more flexible and stress free to complete the game	Wrong number or statistic	Corrected

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5.2 Peer review: watch both assigned LinkedIn videos and score them (0–2 per criterion)

Group reviewed	Problem	Build	Data	Honesty	Clarity	Best moment (mm:ss)	One question for the viva
Group 09: Kush, Himanshi, Nirvan, Jiah	2	2	2	2	1	1:15 min	type here
Group 17: Samarth, Ishan, Aarush, Sachi	2	2	1	2	2	00:30 s	type here

SECTION 6 · Individual Learning Record *not marked · one row per member · answer honestly*

These answers do not change your marks. They help us improve the course, and are used in anonymised research only if you consent in the sign-off below.

Member	Hours I spent (total)	My declared role matched my actual work (1 = not at all, 5 = fully)	Before this course I had used…	Before this course I had tested with real users	I learned a skill from a teammate of another programme	Share of my writing first drafted with AI	Wrong AI facts / citations I personally caught
Varun Iyer (F050)	50	4	Git only	Yes	Yes	26–50 %	1–2
Arjun Salunke (F049)	50	4	Neither Git nor LFS	Yes	Yes	26–50 %	3–5
Om Kadam (F014)	50	5	Neither Git nor LFS	Yes	Yes	0–25 %	1–2

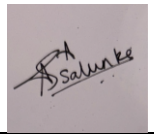
Member	The most valuable thing I learned (one line)	One thing I would change in this course (one line)
Varun Iyer (F050)	I learned how Meta XR can support immersive campus exploration and how XR interactions can help users discover campus spaces.	More Hardware components to be explored
Arjun Salunke (F049)	I learned how to connect Unity scenes, player movement, and interactions so students can explore different campus floors.	More game logic and depth in full fledged development
Om Kadam (F014)	I learned how 3D modelling and world building can turn campus plans into an explorable environment, while keeping unverified details clearly marked.	More designing modules

Track A: how this dossier becomes your 4-page IEEE paper

I. Introduction	Section 1: title, bottleneck, verified standard
II. Related work	Section 1.1: verified sources only (5+ with DOIs)
III. System design	Section 2: architecture, state machine, equations
IV. Evaluation	Section 4: protocol, N, SUS/TLX/metric, plots
V. Discussion & limitations	Sections 3–5: reproducibility (repo, release), AI audit lessons
Target venues (title sheet)	A: IndiaHCI (Indian Conference on Human-Computer Interaction) · B: iLRN (Immersive Learning Research Network)

Academic Sign-off *every member answers and signs*

Answer each statement for yourself: (1) this work is original and every source and AI use is declared; (2) the code in our folder runs and the Release build installs; (3) I understand our whole project and can explain my own part in the viva voce.

Member (Roll)	(1) Original work	(2) Code runs	(3) Viva-ready	Signature	Optional research consent*
Varun Iyer (F050)	Yes	Yes	Yes		Yes
Arjun Salunke (F049)	Yes	Yes	Yes		Yes
Om Kadam (F014)	Yes	Yes	Yes		Yes

**Optional: may Dr. Nanade use anonymised data from this dossier and repository in education research publications? Your answer has no effect on your marks, and you can withdraw until results are published.*